

14.2 DOUBLE INTEGRALS OVER NONRECTANGULAR REGIONS

In this section we will show how to evaluate double integrals over regions other than rectangles.

■ ITERATED INTEGRALS WITH NONCONSTANT LIMITS OF INTEGRATION

Later in this section we will see that double integrals over nonrectangular regions can often be evaluated as iterated integrals of the following types:

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x, y) dy \right] dx \quad (1)$$

$$\int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} f(x, y) dx \right] dy \quad (2)$$

► **Example 1** Evaluate

$$(a) \int_0^1 \int_{-x}^{x^2} y^2 x dy dx \quad (b) \int_0^{\pi/3} \int_0^{\cos y} x \sin y dx dy$$

Solution (a).

$$\begin{aligned} \int_0^1 \int_{-x}^{x^2} y^2 x dy dx &= \int_0^1 \left[\int_{-x}^{x^2} y^2 x dy \right] dx = \int_0^1 \left. \frac{y^3 x}{3} \right|_{y=-x}^{x^2} dx \\ &= \int_0^1 \left[\frac{x^7}{3} + \frac{x^4}{3} \right] dx = \left(\frac{x^8}{24} + \frac{x^5}{15} \right) \Big|_0^1 = \frac{13}{120} \end{aligned}$$

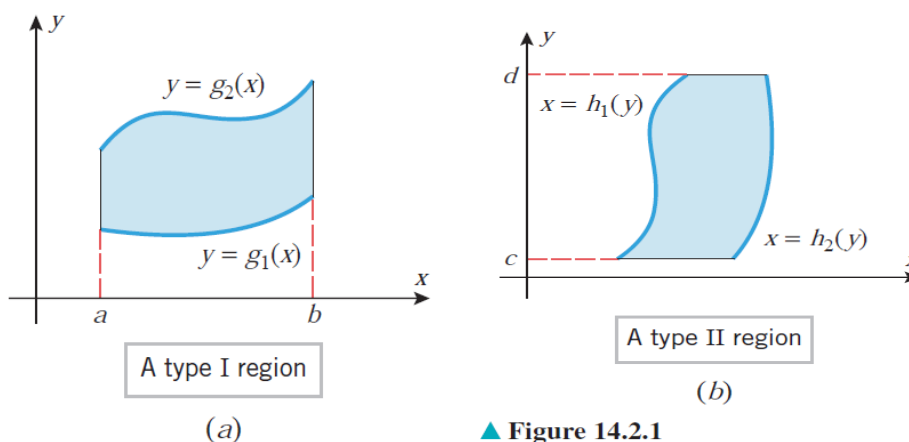
Solution (b).

$$\begin{aligned} \int_0^{\pi/3} \int_0^{\cos y} x \sin y dx dy &= \int_0^{\pi/3} \left[\int_0^{\cos y} x \sin y dx \right] dy = \int_0^{\pi/3} \left. \frac{x^2}{2} \sin y \right|_{x=0}^{\cos y} dy \\ &= \int_0^{\pi/3} \left[\frac{1}{2} \cos^2 y \sin y \right] dy = -\frac{1}{6} \cos^3 y \Big|_0^{\pi/3} = \frac{7}{48} \quad \blacktriangleleft \end{aligned}$$

■ DOUBLE INTEGRALS OVER NONRECTANGULAR REGIONS

14.2.1 DEFINITION

- (a) A **type I region** is bounded on the left and right by vertical lines $x = a$ and $x = b$ and is bounded below and above by continuous curves $y = g_1(x)$ and $y = g_2(x)$, where $g_1(x) \leq g_2(x)$ for $a \leq x \leq b$ (Figure 14.2.1a).
- (b) A **type II region** is bounded below and above by horizontal lines $y = c$ and $y = d$ and is bounded on the left and right by continuous curves $x = h_1(y)$ and $x = h_2(y)$ satisfying $h_1(y) \leq h_2(y)$ for $c \leq y \leq d$ (Figure 14.2.1b).



▲ Figure 14.2.1

14.2.2 THEOREM

- (a) If R is a type I region on which $f(x, y)$ is continuous, then

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx \quad (3)$$

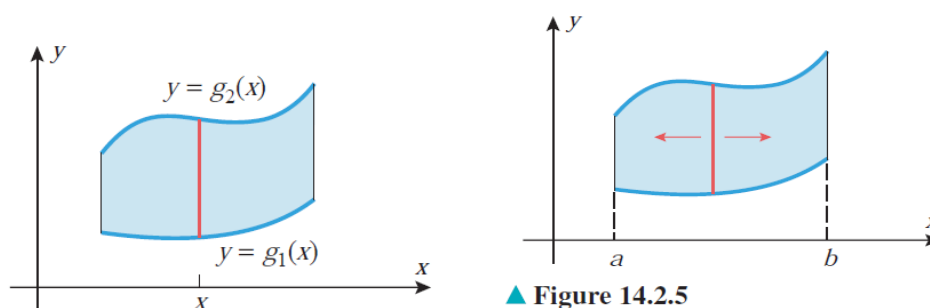
- (b) If R is a type II region on which $f(x, y)$ is continuous, then

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy \quad (4)$$

Determining Limits of Integration: Type I Region

Step 1. Since x is held fixed for the first integration, we draw a vertical line through the region R at an arbitrary fixed value x (Figure 14.2.5). This line crosses the boundary of R twice. The lower point of intersection is on the curve $y = g_1(x)$ and the higher point is on the curve $y = g_2(x)$. These two intersections determine the lower and upper y -limits of integration in Formula (3).

Step 2. Imagine moving the line drawn in Step 1 first to the left and then to the right (Figure 14.2.5). The leftmost position where the line intersects the region R is $x = a$, and the rightmost position where the line intersects the region R is $x = b$. This yields the limits for the x -integration in Formula (3).



▲ Figure 14.2.5

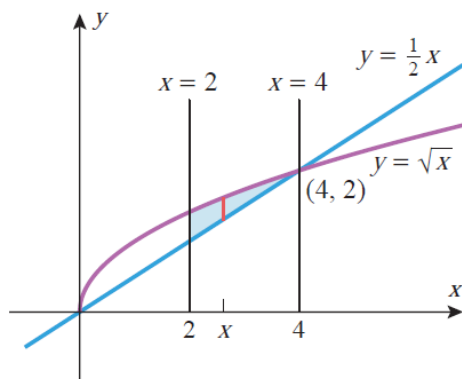
► **Example 3** Evaluate

$$\iint_R xy \, dA$$

over the region R enclosed between $y = \frac{1}{2}x$, $y = \sqrt{x}$, $x = 2$, and $x = 4$.

Solution. We view R as a type I region. The region R and a vertical line corresponding to a fixed x are shown in Figure 14.2.6. This line meets the region R at the lower boundary $y = \frac{1}{2}x$ and the upper boundary $y = \sqrt{x}$. These are the y -limits of integration. Moving this line first left and then right yields the x -limits of integration, $x = 2$ and $x = 4$. Thus,

$$\begin{aligned} \iint_R xy \, dA &= \int_2^4 \int_{x/2}^{\sqrt{x}} xy \, dy \, dx = \int_2^4 \left[\frac{xy^2}{2} \right]_{y=x/2}^{\sqrt{x}} dx = \int_2^4 \left(\frac{x^2}{2} - \frac{x^3}{8} \right) dx \\ &= \left[\frac{x^3}{6} - \frac{x^4}{32} \right]_2^4 = \left(\frac{64}{6} - \frac{256}{32} \right) - \left(\frac{8}{6} - \frac{16}{32} \right) = \frac{11}{6} \quad \blacktriangleleft \end{aligned}$$

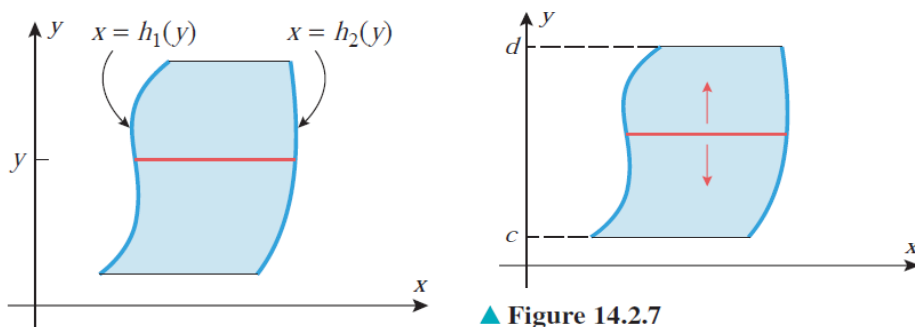


▲ Figure 14.2.6

Determining Limits of Integration: Type II Region

Step 1. Since y is held fixed for the first integration, we draw a horizontal line through the region R at a fixed value y (Figure 14.2.7). This line crosses the boundary of R twice. The leftmost point of intersection is on the curve $x = h_1(y)$ and the rightmost point is on the curve $x = h_2(y)$. These intersections determine the x -limits of integration in (4).

Step 2. Imagine moving the line drawn in Step 1 first down and then up (Figure 14.2.7). The lowest position where the line intersects the region R is $y = c$, and the highest position where the line intersects the region R is $y = d$. This yields the y -limits of integration in (4).



▲ Figure 14.2.7

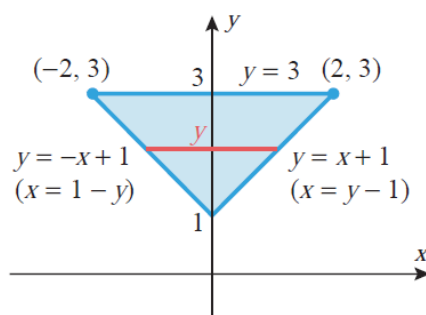
► **Example 4** Evaluate

$$\iint_R (2x - y^2) dA$$

over the triangular region R enclosed between the lines $y = -x + 1$, $y = x + 1$, and $y = 3$.

Solution. We view R as a type II region. The region R and a horizontal line corresponding to a fixed y are shown in Figure 14.2.8. This line meets the region R at its left-hand boundary $x = 1 - y$ and its right-hand boundary $x = y - 1$. These are the x -limits of integration. Moving this line first down and then up yields the y -limits, $y = 1$ and $y = 3$. Thus,

$$\begin{aligned}\iint_R (2x - y^2) dA &= \int_1^3 \int_{1-y}^{y-1} (2x - y^2) dx dy = \int_1^3 [x^2 - y^2 x]_{x=1-y}^{y-1} dy \\ &= \int_1^3 [(1 - 2y + 2y^2 - y^3) - (1 - 2y + y^3)] dy \\ &= \int_1^3 (2y^2 - 2y^3) dy = \left[\frac{2y^3}{3} - \frac{y^4}{2} \right]_1^3 = -\frac{68}{3} \quad \blacktriangleleft\end{aligned}$$



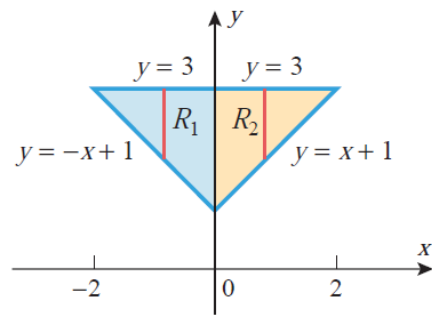
▲ Figure 14.2.8

Observations:

In Example 4 we could have treated R as a type I region, but with an added complication. Viewed as a type I region, the upper boundary of R is the line $y = 3$ (Figure 14.2.9) and the lower boundary consists of two parts, the line $y = -x + 1$ to the left of the y -axis and the line $y = x + 1$ to the right of the y -axis. To carry out the integration it is necessary to decompose the region R into two parts, R_1 and R_2 , as shown in Figure 14.2.9, and write

$$\begin{aligned}\iint_R (2x - y^2) dA &= \iint_{R_1} (2x - y^2) dA + \iint_{R_2} (2x - y^2) dA \\ &= \int_{-2}^0 \int_{-x+1}^3 (2x - y^2) dy dx + \int_0^2 \int_{x+1}^3 (2x - y^2) dy dx\end{aligned}$$

This will yield the same result that was obtained in previous example (Example 4).



▲ Figure 14.2.9

Practice Problem:

Chapter-14.2:

Problems: 3, 4, 5, 8, 15-26

Thank you for your attention!